

Karthika Vellingiri

Professional Summary

Data analytics professional with over 11 years of experience in telecom billing systems and operational data analysis. Skilled in SQL, Python, and data visualization tools with expertise in data validation, root cause analysis, and reporting automation. Currently pursuing a Master's in Data Science at Bellevue University, applying machine learning and statistical analysis to real-world datasets across healthcare, retail, and telecom domains.

Technical Skills

Programming: Python, SQL, PL/SQL

Data Analysis: Pandas, NumPy, Data Cleaning, Statistical Analysis

Machine Learning: Scikit-learn, Regression, Classification, Random Forest

Data Visualization: Tableau, Power BI, MicroStrategy, Matplotlib, Seaborn

Tools: Jupyter Notebook, Git, GitHub

Professional Experience

Telecom Billing Analyst / Business Analyst – Telecommunications Industry

- Analyzed high-volume telecom billing data to identify discrepancies and improve billing accuracy.
- Performed data validation and root cause analysis to resolve operational and billing issues.
- Developed operational reports and dashboards to support business monitoring and decision-making.
- Collaborated with cross-functional teams to gather requirements and implement analytical solutions.
- Supported reporting automation initiatives that improved operational efficiency.

Education

Master of Science in Data Science – Bellevue University, Nebraska (Expected 2026)

Bachelor's Degree – Anna University, India

Data Science Projects

- AI-Driven Retail Optimization – Developed predictive models to forecast product demand and analyze pricing strategies.
- Smart Healthcare Operations – Built models to forecast patient demand and optimize staff allocation.
- Telecom Customer Churn Prediction – Developed classification models to identify customers at risk of churn.
- Childcare Cost Affordability Analysis – Created Tableau dashboards analyzing childcare cost trends across the United States.
- Weather Data Retrieval Application – Built a Python application to retrieve real-time weather data using API integration.

Portfolio

GitHub: <https://github.com/Karthika-MSDS>